

Study Guide — Lecture 17

The Higgs Mechanism in Unitary Gauge & the Lepton Standard Model

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Study Guide — Lecture 17: Higgs Mechanism & the Lepton Standard Model

Part I Core Concepts and Physical Intuition

1.1 The Central Problem: Gauge Invariance vs. Mass

The gauge-invariance principle demands massless gauge bosons. For a local $U(1)$ symmetry, any term $\frac{1}{2}m^2 A_\mu A^\mu$ is forbidden because under $A_\mu \rightarrow A_\mu - \frac{1}{g}\partial_\mu\theta$ it transforms to something that is not gauge-invariant. Yet the $W^\pm$ and Z bosons are experimentally massive ($m_W \approx 80$ GeV, $m_Z \approx 91$ GeV). The resolution is that the **vacuum**, not the Lagrangian, breaks the symmetry.

Physical intuition. The Lagrangian is gauge-invariant, but the ground state is not. A field that condenses ($\langle\phi\rangle \neq 0$) selects a preferred direction in field space, hiding the symmetry without violating it in the equations of motion. The resulting physics is distinct from an explicit mass term, and the two situations are experimentally distinguishable by the constrained relationship among all couplings and masses that SSB predicts.

1.2 The Role of the Local vs. Global Symmetry

- For a **global** broken symmetry (Goldstone's theorem): each broken generator produces a massless scalar — the Goldstone boson. These particles remain in the physical spectrum.
- For a **local** broken symmetry (Higgs mechanism): the transformation parameter $\theta(x)$ is a function of spacetime. This extra freedom allows us to *rotate the Goldstone field away point by point* — something a global transformation cannot do. The Goldstone boson is **eaten** by the gauge field, supplying its missing longitudinal polarization and thereby generating mass.

The critical word is **local**: the gauge freedom is what licenses the complete removal of the would-be Goldstone from the physical spectrum.

1.3 Three Universal Lessons for Any Local SSB

Physical Law 0.1 Universal Lessons of Local SSB

The following hold for every gauge group G broken spontaneously to a subgroup H , regardless of whether G is Abelian, non-Abelian, or a product group:

1. **Broken generators $\Rightarrow$ massive gauge bosons.** Each generator broken by the vev gives mass to one gauge boson. There are $\dim G - \dim H$ broken generators, hence that many massive vectors.
2. **Unbroken generators $\Rightarrow$ massless gauge bosons.** The gauge bosons associated with the $\dim H$ surviving generators remain exactly massless, *protected* by the

residual symmetry.

3. **The condensing field must be a scalar.** A field carrying non-trivial Lorentz quantum numbers (spinor, vector) would break Lorentz invariance upon condensing. Only a Lorentz-scalar vev is consistent with the observed Lorentz invariance of the vacuum.

Part II Rigorous Mathematical Derivations

2.1 The Abelian Higgs Model: Setup

Consider a complex scalar ϕ of charge $+1$ under a local $U(1)$, with gauge-invariant Lagrangian

$$\mathcal{L} = (D_\mu \phi)^\dagger D^\mu \phi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \mu^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2, \quad D_\mu \phi = (\partial_\mu + ig A_\mu) \phi. \quad (1)$$

For $\mu^2 < 0$ the potential has a ring of minima at

$$|\langle \phi \rangle| = \frac{v}{\sqrt{2}}, \quad v^2 = -\frac{\mu^2}{\lambda} > 0. \quad (2)$$

2.2 The Nonlinear (Magnitude–Phase) Realization

Instead of the linear split $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$, write ϕ in polar field variables:

$$\phi(x) = e^{i\chi(x)/v} \frac{v + h(x)}{\sqrt{2}}, \quad (3)$$

where $h(x)$ is the radial (Higgs) excitation and $\chi(x)$ is the angular (Goldstone) excitation, both with vanishing vev. This is the **nonlinear realization**.

Why this parameterization? It isolates the gauge-transformable phase χ as a multiplicative exponential, making the unitary gauge choice transparent: a single local $U(1)$ transformation can eliminate $\chi(x)$ everywhere simultaneously.

2.3 The Unitary Gauge: Eliminating χ

The theory (1) is still locally $U(1)$ -invariant even after SSB (the *equations of motion* respect the symmetry; only the vacuum does not). Any local transformation $\phi \rightarrow e^{i\theta(x)} \phi$ is a valid redescription. Choose

$$\theta(x) = -\frac{\chi(x)}{v} \quad (\text{unitary gauge}). \quad (4)$$

Under this transformation the fields become:

$$\phi \rightarrow \phi' = \frac{v + h(x)}{\sqrt{2}} \quad (\text{real, positive}), \quad (5)$$

$$A_\mu \rightarrow V_\mu = A_\mu - \frac{1}{g} \partial_\mu \theta = A_\mu + \frac{1}{gv} \partial_\mu \chi. \quad (6)$$

The field χ has not been deleted — it has been *transferred* into the gauge field V_μ . It will reappear as the longitudinal polarization of the now-massive vector.

2.4 Expanding the Lagrangian in Unitary Gauge

Step 1: Covariant derivative on the real field.

$$D_\mu \phi' = (\partial_\mu + igV_\mu) \frac{v+h}{\sqrt{2}} = \frac{1}{\sqrt{2}} \partial_\mu h + \frac{ig}{\sqrt{2}} V_\mu (v+h). \quad (7)$$

The modulus squared is

$$|D_\mu \phi'|^2 = \frac{1}{2} (\partial_\mu h)^2 + \frac{g^2}{2} V_\mu V^\mu (v+h)^2. \quad (8)$$

Step 2: Scalar potential in shifted form. Rewrite $V(\phi) = \lambda(\phi^\dagger \phi - v^2/2)^2$ (equal to the original up to a constant). With $\phi'^\dagger \phi' = (v+h)^2/2$:

$$\phi'^\dagger \phi' - \frac{v^2}{2} = vh + \frac{h^2}{2} \implies V = \lambda \left(vh + \frac{h^2}{2} \right)^2 = \lambda v^2 h^2 + \lambda v h^3 + \frac{\lambda}{4} h^4. \quad (9)$$

Step 3: Expand $(v+h)^2 = v^2 + 2vh + h^2$ in (8) and collect all terms.

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} (\partial_\mu h)^2 + \underbrace{\frac{1}{2} g^2 v^2 V_\mu V^\mu}_{\text{vector mass}} - \underbrace{\frac{1}{2} (2\lambda v^2) h^2}_{\text{scalar mass}} + \frac{g^2}{2} V_\mu V^\mu h(2v+h) - \lambda v h^3 - \frac{\lambda}{4} h^4, \quad (10)$$

where $F_{\mu\nu}$ is now built from V_μ . **The field χ is absent.**

2.5 Reading Off the Physical Spectrum

Quadratic terms in (10) determine the masses. For a *real* massive vector the mass term is $+\frac{1}{2} m_V^2 V_\mu V^\mu$; for a real scalar it is $-\frac{1}{2} m_h^2 h^2$. Reading off:

$$\boxed{m_V^2 = g^2 v^2, \quad m_h^2 = 2\lambda v^2.} \quad (11)$$

The two physical particles of the model are therefore a **massive vector boson** V_μ and a **massive real scalar** h (the Higgs boson). Both masses vanish as $v \rightarrow 0$, confirming the vev as their sole source.

2.6 Degree-of-Freedom Bookkeeping (Abelian Case)

Field / Mode	Before SSB	After SSB
Gauge field A_μ (massless)	2 transverse	—
Phase χ (real scalar)	1	—
Radial h (real scalar)	1	1 (Higgs boson)
Vector V_μ (massive)	—	3 (2 transverse + 1 longitudinal)
Total	4	4

In symbols: $\underbrace{2}_{A_\mu \text{ massless}} + \underbrace{1}_{\chi} = \underbrace{3}_{V_\mu \text{ massive}}$, plus 1 for h on both sides. No degrees of freedom are created or destroyed; χ becomes the longitudinal polarization of V_μ .

2.7 Coupling Relations from SSB

All interaction terms in (10) descend from only **three** parameters $\{\lambda, v, g\}$. Two structural facts:

1. **hVV coupling** $\propto m_V^2/v$: The term $\frac{g^2}{2}V_\mu V^\mu(2vh)$ has coefficient $g^2v = m_V^2/v$:

$$g_{hVV} = g^2v = \frac{m_V^2}{v}. \quad (12)$$

The Higgs boson couples most strongly to the heaviest gauge bosons — the LHC discovery handle.

2. **Dimensionless couplings are preserved by SSB:** The h^4 coefficient ($\lambda/4$) and the h^2V^2 coefficient ($g^2/2$) are dimensionless and match their values in the symmetric Lagrangian. Only operators with positive mass-dimension coefficients are shifted by the vev. This is a standard sanity check after any SSB calculation.

Note:

Normalization pitfall: From $\frac{1}{2}g^2v^2V_\mu V^\mu$ one reads $m_V^2 = g^2v^2$, i.e. $m_V = gv$ — *no* extra $\sqrt{2}$. But for a Dirac fermion, $m_\psi = yv/\sqrt{2}$ (the $\sqrt{2}$ comes from the vev convention $\langle\phi\rangle = v/\sqrt{2}$), and for the scalar, $m_h^2 = 2\lambda v^2$ (the factor of 2 comes from $\lambda(vh + h^2/2)^2$ expanded). Confusing these three normalizations is the most common error in spectrum questions.

2.8 Algorithmic Method for Any Local SSB

1. Identify gauge group G , matter representations, and write the gauge-invariant Lagrangian.
2. Minimize the potential; record $\langle\phi\rangle$ and the scale v .
3. **Test generators on the vacuum:** $T^a\langle\phi\rangle = 0$ means T^a is unbroken; otherwise it is broken. The unbroken generators form $H \subset G$.
4. Count: $\dim G - \dim H$ broken generators $\Rightarrow$ that many would-be Goldstones eaten $\Rightarrow$ that many gauge bosons become massive.
5. Go to unitary gauge (rotate Goldstone phases to zero) and expand $|D_\mu\phi|^2$ around the vev: terms quadratic in gauge fields are the mass matrix.
6. Expand the potential around the vev for scalar masses; verify DoF balance.

2.9 Fermion Masses in the Unitary Gauge

Add chiral fermions ψ_L, ψ_R with charges satisfying the gauge-invariance condition

$$q(\psi_R) - q(\psi_L) = q(\phi). \quad (13)$$

This ensures the Yukawa term $-y\phi\bar{\psi}_R\psi_L + \text{h.c.}$ is gauge-invariant while a bare mass $m\bar{\psi}_R\psi_L$ is forbidden. In unitary gauge $\phi \rightarrow (v+h)/\sqrt{2}$:

$$\mathcal{L}_{\text{Yuk}} = - \underbrace{\frac{yv}{\sqrt{2}}}_{m_\psi} \bar{\psi}_R\psi_L - \frac{y}{\sqrt{2}} h \bar{\psi}_R\psi_L + \text{h.c.} \quad (14)$$

The first term is a Dirac mass $m_\psi = yv/\sqrt{2}$; the second gives the Higgs–fermion coupling $g_{h\bar{\psi}\psi} = m_\psi/v$. Mass and Yukawa coupling are therefore determined by the *same* parameter — a falsifiable prediction of the mechanism.

2.10 The Lepton Standard Model: Gauge Structure

Physical Concept 0.1: The Lepton Standard Model (LSM)

The LSM is the gauge theory with local symmetry $G_{\text{LSM}} = SU(2)_L \times U(1)_Y$ (hypercharge, *not* electromagnetism), field content as in Table 1, and spontaneous breaking pattern

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{EM}}. \quad (15)$$

The full Standard Model is the LSM plus quarks and $SU(3)_c$ (QCD).

Field	$SU(2)_L$ rep	$U(1)_Y$ charge Y	Description
$L_{Li} = \begin{pmatrix} \nu_{Li} \\ e_{Li} \end{pmatrix}$	2	$-\frac{1}{2}$	Left-handed lepton doublet, $i = 1, 2, 3$
E_{Ri}	1	-1	Right-handed charged lepton, $i = 1, 2, 3$
$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}$	2	$+\frac{1}{2}$	Complex Higgs doublet

Table 1: Field content of the Lepton Standard Model. Nine chiral (Weyl) fermions in two kinds of representations: 3×2 from doublets plus 3×1 from singlets.

Gauge sector. The group $SU(2)_L \times U(1)_Y$ has four generators: $\{T^a\}_{a=1,2,3}$ for $SU(2)_L$ and Y for $U(1)_Y$, with algebra

$$[T^a, T^b] = i\epsilon^{abc}T^c, \quad [T^a, Y] = 0. \quad (16)$$

The two factors are independent, hence the commutator of generators from different factors vanishes.

Note:

Exam trap — number of gauge couplings: The number of independent gauge couplings equals the number of *simple or* $U(1)$ *factors* in the gauge group, *not* the number of generators. $SU(2)_L \times U(1)_Y$ has **two** independent couplings: g (for $SU(2)_L$) and g' (for $U(1)_Y$).

Locality requires one gauge boson per generator, four total:

$$W_\mu^a \sim (\mathbf{3})_0, \quad a = 1, 2, 3 \quad (\text{adjoint of } SU(2)_L, \text{ zero hypercharge}), \quad (17)$$

$$B_\mu \sim (\mathbf{1})_0 \quad (\text{singlet of } SU(2)_L, \text{ zero hypercharge}). \quad (18)$$

Field strengths.

$$W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a - g\epsilon^{abc}W_\mu^b W_\nu^c, \quad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (19)$$

The non-Abelian $W_{\mu\nu}^a$ carries a cubic term in the structure constants (giving rise to 3- and 4-boson self-interactions); the Abelian $B_{\mu\nu}$ does not.

2.11 Covariant Derivatives for Each LSM Field

The general covariant derivative is

$$D_\mu = \partial_\mu + igW_\mu^a T^a + ig'Y B_\mu, \quad (20)$$

where $T^a = \sigma^a/2$ on $SU(2)$ doublets and $T^a = 0$ on singlets, and Y is the field's hypercharge eigenvalue. Explicitly:

$$D_\mu \phi = \left(\partial_\mu + \frac{ig}{2} W_\mu^a \sigma^a + \frac{ig'}{2} B_\mu \right) \phi, \quad (21)$$

$$D_\mu L_L = \left(\partial_\mu + \frac{ig}{2} W_\mu^a \sigma^a - \frac{ig'}{2} B_\mu \right) L_L, \quad (22)$$

$$D_\mu E_R = (\partial_\mu - ig' B_\mu) E_R. \quad (23)$$

The differences trace directly to the representation and hypercharge of each field ($Y = +\frac{1}{2}, -\frac{1}{2}, -1$ for ϕ, L_L, E_R respectively). The $SU(2)$ piece vanishes for the singlet E_R .

2.12 No Bare Lepton Mass Term

A Lorentz-invariant, renormalizable fermion mass requires the combination $\bar{L}_L E_R$. This fails *both* gauge tests:

- $SU(2)_L$: $\bar{L}_L$ is a doublet ($\bar{\mathbf{2}}$) and E_R is a singlet ($\mathbf{1}$), so $\bar{L}_L E_R$ transforms as a doublet — not gauge-invariant.
- $U(1)_Y$: Hypercharges add to $+\frac{1}{2} + (-1) = -\frac{1}{2} \neq 0$ — not invariant.

$$\boxed{\text{No bare lepton mass is permitted by the gauge representations.}} \quad (24)$$

This is a theorem, not an assumption. Every observed lepton mass must arise from SSB.

2.13 The Yukawa Sector of the LSM

The unique renormalizable gauge-invariant coupling between leptons and the Higgs is

$$\boxed{\mathcal{L}_{\text{Yuk}} = -Y_{ij}^e \bar{L}_{Li} \phi E_{Rj} + \text{h.c.}} \quad (25)$$

Invariance check.

- $U(1)_Y$: $Y(\bar{L}_L) + Y(\phi) + Y(E_R) = +\frac{1}{2} + \frac{1}{2} + (-1) = 0$. ✓
- $SU(2)_L$: $\bar{\mathbf{2}} \otimes \mathbf{2} \supset \mathbf{1}$ — the two doublets contract to a singlet. ✓

The coupling Y_{ij}^e is a 3×3 complex matrix of dimensionless numbers ($2 \times \frac{3}{2} + 1 = 4$, leaving dimension-zero coefficient). A basis change among generations diagonalizes it:

$$Y^e = \text{diag}(y_e, y_\mu, y_\tau) \quad (\text{mass basis}). \quad (26)$$

Once the Higgs acquires its vev these three eigenvalues become the electron, muon, and tau masses: $m_\ell = y_\ell v / \sqrt{2}$.

2.14 The Scalar Sector and Choice of Vacuum

The potential is the same Mexican-hat form

$$V(\phi) = -\mu^2 \phi^\dagger \phi - \lambda(\phi^\dagger \phi)^2 = \lambda \left(\phi^\dagger \phi - \frac{v^2}{2} \right)^2 + \text{const}, \quad v^2 = -\frac{\mu^2}{\lambda}. \quad (27)$$

For $\mu^2 < 0$ the doublet condenses with $|\langle \phi \rangle| = v/\sqrt{2}$. The potential fixes only the *magnitude*; the *direction* in the doublet is our convention. We place the vev in the real part of the lower ($T^3 = -\frac{1}{2}$) component:

$$\boxed{\langle \phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}}. \quad (28)$$

Why the lower component? We need the condensing component to be electrically neutral — a charged vacuum would spontaneously break $U(1)_{\text{EM}}$, giving the photon a mass. We verify below that the lower component of ϕ carries $Q = 0$.

2.15 Deriving the Unbroken Generator: Electric Charge

A generator Q is unbroken iff $Q\langle \phi \rangle = 0$. We test all four generators $\{T^1, T^2, T^3, Y\}$ on the vev (28).

T^1 and T^2 : $\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ are off-diagonal, so

$$T^1 \langle \phi \rangle = \frac{\sigma^1}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = \frac{1}{2\sqrt{2}} \begin{pmatrix} v \\ 0 \end{pmatrix} \neq 0, \quad T^2 \langle \phi \rangle = \frac{\sigma^2}{2} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = \frac{i}{2\sqrt{2}} \begin{pmatrix} v \\ 0 \end{pmatrix} \neq 0. \quad (29)$$

Both T^1 and T^2 are **broken**.

T^3 and Y :

$$T^3 \langle \phi \rangle = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = -\frac{1}{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad (30)$$

$$Y \langle \phi \rangle = +\frac{1}{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \quad (31)$$

These are equal in magnitude and opposite in sign. Their sum therefore vanishes:

$$\boxed{Q \equiv T^3 + Y, \quad Q \langle \phi \rangle = \left(-\frac{1}{2} + \frac{1}{2} \right) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = 0.} \quad (32)$$

This single linear combination $Q = T^3 + Y$ is the **unbroken generator**. We identify $U(1)_Q$ with $U(1)_{\text{EM}}$ and Q with the electric charge operator.

Physical intuition. Electric charge is not imposed on the Standard Model by hand. It *emerges* as the unique combination of weak isospin and hypercharge that the Higgs vacuum happens to preserve. Had we placed the vev in the upper component ($T^3 = +\frac{1}{2}$), the preserved combination would be $Q = T^3 - Y$ — same physics, different convention.

Note:

Hypercharge convention: The formula $Q = T^3 + Y$ uses the normalization where the lepton doublet has $Y_L = -\frac{1}{2}$. Many textbooks (e.g. Peskin & Schroeder) normalize hypercharge

by a factor of 2: $\tilde{Y}_L = -1$, giving $Q = T^3 + \tilde{Y}/2$. Always check the convention; note also that for $SU(2)$ singlets $T^3 = 0$, so a singlet's electric charge *equals* its hypercharge.

2.16 Electric Charges of All LSM Fields

Worked Example 0.1 (Electric Charges via $Q = T^3 + Y$)

For doublets $T^3 = \pm \frac{1}{2}$ (upper/lower component); for singlets $T^3 = 0$.

$$\begin{aligned} Q(\nu_L) &= +\frac{1}{2} + (-\frac{1}{2}) = 0 && \text{(neutrino — neutral fermion)} \\ Q(e_L) &= -\frac{1}{2} + (-\frac{1}{2}) = -1 && \text{(left-handed electron)} \\ Q(e_R) &= 0 + (-1) = -1 && \text{(right-handed electron)} \\ Q(\phi^+) &= +\frac{1}{2} + \frac{1}{2} = +1 && \text{(charged Higgs component)} \\ Q(\phi^0) &= -\frac{1}{2} + \frac{1}{2} = 0 && \text{(neutral Higgs component — carries the vev)} \end{aligned}$$

Both chiralities of the electron have $Q = -1$, making the electron vector-like under $U(1)_{\text{EM}}$ — a necessary condition for SSB to deliver a fermion mass. The vev component ϕ^0 is electrically neutral as required.

2.17 The Higgs Doublet in Unitary Gauge

Four generators, one unbroken $\Rightarrow$ three broken $\Rightarrow$ three would-be Goldstones eaten. Of the four real degrees of freedom of the complex doublet, three disappear and one physical scalar survives:

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \quad (\text{unitary gauge}). \quad (33)$$

The surviving field h is the **Higgs boson**: a real, electromagnetically neutral, color-singlet scalar with

$$m_h^2 = 2\lambda v^2, \quad m_h^{\text{measured}} = 125.09 \pm 0.24 \text{ GeV} \quad (\text{LHC 2012, Nobel 2013}). \quad (34)$$

2.18 Gauge-Boson Mass Matrix: Full Derivation

The gauge-boson masses come from $|D_\mu \langle \phi \rangle|^2$. Setting $\phi = \langle \phi \rangle$ and dropping ∂_μ (which annihilates a constant):

$$D_\mu \langle \phi \rangle = \frac{i}{2} (gW_\mu^a \sigma^a + g'B_\mu) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = \frac{i}{\sqrt{8}} (gW_\mu^a \sigma^a + g'B_\mu) \begin{pmatrix} 0 \\ v \end{pmatrix}. \quad (35)$$

Step 1: Expand the 2×2 matrix. Using $\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and $B_\mu \cdot \mathbf{1}_2$:

$$gW_\mu^a \sigma^a + g'B_\mu \mathbf{1}_2 = \begin{pmatrix} gW_\mu^3 + g'B_\mu & g(W_\mu^1 - iW_\mu^2) \\ g(W_\mu^1 + iW_\mu^2) & -gW_\mu^3 + g'B_\mu \end{pmatrix}. \quad (36)$$

Step 2: Act on the vev vector. Acting (36) on $(0, v)^T$ selects the second column:

$$(gW_\mu^a \sigma^a + g'B_\mu) \begin{pmatrix} 0 \\ v \end{pmatrix} = v \begin{pmatrix} g(W_\mu^1 - iW_\mu^2) \\ -gW_\mu^3 + g'B_\mu \end{pmatrix}. \quad (37)$$

Step 3: Form the modulus squared.

$$\begin{aligned}
\mathcal{L}_{\text{mass}} &= (D_\mu \langle \phi \rangle)^\dagger D^\mu \langle \phi \rangle \\
&= \frac{v^2}{8} \left[g^2 (W_\mu^1)^2 + g^2 (W_\mu^2)^2 + (gW_\mu^3 - g'B_\mu)^2 \right] + \frac{v^2}{8} |g(W_\mu^1 - iW_\mu^2)|^2_{(\text{same as above})} \\
&= \frac{v^2}{8} \left[g^2 (W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu}) + (gW_\mu^3 - g'B_\mu)(gW^{3\mu} - g'B^\mu) \right].
\end{aligned} \tag{38}$$

$$\boxed{\mathcal{L}_{\text{mass}} = \frac{v^2}{8} \left[g^2 (W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu}) + (gW_\mu^3 - g'B_\mu)(gW^{3\mu} - g'B^\mu) \right].} \tag{39}$$

Step 4: Identify the mass eigenstates. **Charged sector** (W^1, W^2): Already diagonal. Define the complex (physical) charged bosons

$$W_\mu^\pm = \frac{W_\mu^1 \mp iW_\mu^2}{\sqrt{2}}, \tag{40}$$

so $W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu} = 2W_\mu^+ W^{-\mu}$. Reading off the mass:

$$\frac{v^2}{8} \cdot g^2 \cdot 2W^+ W^- = \frac{g^2 v^2}{4} W^+ W^- \implies m_W^2 = \frac{g^2 v^2}{4}, \quad m_W = \frac{gv}{2}. \tag{41}$$

Neutral sector (W^3, B): The combination $gW_\mu^3 - g'B_\mu$ appears with coefficient $\frac{v^2}{8}$. The orthogonal combination $g'W_\mu^3 + gB_\mu$ (proportional to the photon A_μ) is *absent*: it was annihilated by Q . The neutral mass matrix is

$$\frac{v^2}{8} \begin{pmatrix} W^3 & B \end{pmatrix} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} \begin{pmatrix} W^3 \\ B \end{pmatrix}. \tag{42}$$

This 2×2 matrix has eigenvalues 0 (photon) and $\frac{v^2}{4}(g^2 + g'^2)$ (the Z boson). Diagonalization is achieved by the **Weinberg angle** θ_W (to be derived in Lecture 18):

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \sin \theta_W & \cos \theta_W \\ \cos \theta_W & -\sin \theta_W \end{pmatrix} \begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix}, \quad \tan \theta_W = \frac{g'}{g}, \tag{43}$$

giving

$$m_A = 0, \quad m_Z^2 = \frac{(g^2 + g'^2)v^2}{4} = \frac{m_W^2}{\cos^2 \theta_W}. \tag{44}$$

Note:

Pitfall: In (39) the combination $g'W_\mu^3 + gB_\mu$ (the photon direction, un-normalized) never appears. Do not expect four eigenvalues in the neutral sector — the photon mass is zero by construction, protected by the unbroken $U(1)_{\text{EM}}$. If your computed neutral mass matrix has two nonzero eigenvalues, check the vev direction or the sign of the $U(1)_Y$ covariant derivative.

Part III Strategic Summary and Degree-of-Freedom Ledger

3.1 Full DoF Bookkeeping for the LSM

State	DoF before SSB	DoF after SSB
Complex Higgs doublet ϕ (4 real d.o.f.)	4	1 (Higgs boson h)
3 eaten Goldstones	3	0
Massless $W_\mu^{1,2,3}, B_\mu$ (2 each)	$4 \times 2 = 8$	0
Massive $W_\mu^\pm, Z_\mu$ (3 each)	0	$3 \times 3 = 9$
Massless photon A_μ (2)	0	2
Total	12	12

In formula: $4 + 8 = 12 = 9 + 2 + 1$.

3.2 Master Summary of the SSB Mechanism

Physical Law 0.2 Structural Summary of the Higgs Mechanism

Given a gauge theory with group G broken spontaneously to H by a scalar vev:

1. $\dim G - \dim H$ gauge bosons eat the would-be Goldstones and become **massive**; $\dim H$ gauge bosons remain **massless**.
2. All masses (vector, scalar, fermion) are proportional to the vev v and vanish as $v \rightarrow 0$.
3. The Higgs boson is the *radial* excitation of the condensing field, with $m_h^2 = 2\lambda v^2$.
4. Yukawa couplings y_f generate fermion masses $m_f = y_f v / \sqrt{2}$ and predict Higgs–fermion couplings $g_{hff} = m_f / v$ — a direct observational test.
5. The hVV coupling is $g_{hVV} = m_V^2 / v$: the Higgs talks most strongly to the heaviest gauge bosons.
6. Dimensionless couplings (e.g. λ, g, g') are unchanged by SSB; only operators with positive mass-dimension coefficients are shifted by v .
7. Electric charge $Q = T^3 + Y$ is the *emergent* unbroken generator, not a fundamental input.

3.3 Critical Pitfalls and Normalization Reference

Note:

Three-normalization rule. Keep these three distinct:

- Real vector: $\mathcal{L} \supset +\frac{1}{2}m_V^2 V_\mu V^\mu \Rightarrow m_V = gv$.
- Dirac fermion: $\mathcal{L} \supset -m_f \bar{\psi}\psi \Rightarrow m_f = yv / \sqrt{2}$.
- Real scalar: $\mathcal{L} \supset -\frac{1}{2}m_h^2 h^2 \Rightarrow m_h^2 = 2\lambda v^2$.

The factors of $\sqrt{2}$ and 2 come from the canonical normalizations of different spin fields, not from SSB itself.

Quantity	Correct formula	Common error
Real vector mass	$m_V^2 = g^2 v^2$ (from $\frac{1}{2}m_V^2 V^2$)	Extra $\sqrt{2}$
Charged W mass	$m_W = gv/2$ (doublet vev)	Writing gv
Dirac fermion mass	$m_f = y_f v/\sqrt{2}$	yv or $yv/2$
Higgs boson mass	$m_h^2 = 2\lambda v^2$ (not λv^2)	Factor of 2 missing
Gauge couplings	g (for $SU(2)_L$), g' (for $U(1)_Y$) — two, not four	Confusing couplings with generators
Q formula	$Q = T^3 + Y$ (with $Y_L = -\frac{1}{2}$ convention)	$Q = T^3 + Y/2$ (P&S convention)
Photon mass	0 (automatic, no fourth eigenvalue)	Looking for four neutral eigenstates

Table 2: Quick-reference normalization and formula table. The mass formulae differ by spin because the canonical kinetic-term normalizations differ.

3.4 Physical Analogy: The Meissner Effect

In a superconductor the ground state (a Cooper-pair condensate) breaks the electromagnetic $U(1)$. By the same mechanism derived above, the photon inside the material acquires an effective mass m_γ^{eff} . A massive photon cannot sustain long-range fields: electromagnetic fields are expelled except within a skin depth $\sim 1/m_\gamma^{\text{eff}}$ (the London penetration length). This is the **Meissner effect** — literally the Higgs mechanism realized in condensed matter, decades before the electroweak theory.

Appendix: Worked Generator Action on the LSM Vev

For reference, here is the complete action of each generator of $SU(2)_L \times U(1)_Y$ on the vev $\langle\phi\rangle = \frac{1}{\sqrt{2}}(0, v)^T$, using $T^a = \sigma^a/2$ and $Y_\phi = +\frac{1}{2}$:

$$T^1\langle\phi\rangle = \frac{1}{2\sqrt{2}}\begin{pmatrix} v \\ 0 \end{pmatrix} \neq 0 \quad \Rightarrow \quad T^1 \text{ broken}, \quad (45)$$

$$T^2\langle\phi\rangle = \frac{i}{2\sqrt{2}}\begin{pmatrix} v \\ 0 \end{pmatrix} \neq 0 \quad \Rightarrow \quad T^2 \text{ broken}, \quad (46)$$

$$T^3\langle\phi\rangle = -\frac{1}{2\sqrt{2}}\begin{pmatrix} 0 \\ v \end{pmatrix} \neq 0 \quad \Rightarrow \quad T^3 \text{ broken}, \quad (47)$$

$$Y\langle\phi\rangle = +\frac{1}{2\sqrt{2}}\begin{pmatrix} 0 \\ v \end{pmatrix} \neq 0 \quad \Rightarrow \quad Y \text{ broken}, \quad (48)$$

$$(T^3 + Y)\langle\phi\rangle = \mathbf{0} \quad \Rightarrow \quad Q = T^3 + Y \text{ unbroken}. \quad (49)$$

The cancellation in the last line is the algebraic origin of electromagnetism.